3. Statistical Estimation and Hypothesis Testing

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MTH345: Deep Generative Models: A Statistical Perspective

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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- We have previously formulated basic probability distributions. This lecture reviews how to find optimal distributions using likelihoods, as well as the statistical testing frameworks based on them.
- Useful materials include:
  - Chapters 8 and 9 from Mathematical Statistics and Data Analysis [MSDA] (2)
- Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

Likelihood

- Consider a random experiment of tossing a potentially unfair coin 10 times.
- Let $\theta = p$ denote the probability of getting heads, and suppose we observe the following sequence:

$$\text{H, T, T, H, H, T, H, T, H, H} \tag{1}$$

- When $\theta = 0.5$ (a fair coin), the corresponding probability table is given by:

Outcome (x)	Tail (0)	Head (1)
$P_{\theta=0.5}(X = x)$	50%	50%

- Based on this assumption, the probability of observing this specific sequence is:

$$P_{\theta=0.5}(\text{HTTHHTHTHH}) = (0.5)^6(0.5)^4 \approx 0.0009766$$

Likelihood

- Let's consider another case with $\theta = 0.7$ (an unfair coin). Then, the corresponding probability table is given by:

Outcome (x)	Tail (0)	Head (1)
$P_{\theta=0.7}(X = x)$	30%	70%

- Based on this assumption, the probability of observing this specific sequence is:

$$P_{\theta=0.7}(\text{HTTHHTHTHH}) = (0.7)^6(0.3)^4 \approx 0.0009530$$

- The probability is decreased from 0.0009766 to 0.0009530.
- **Q:** Which distribution P_θ would maximize $P_\theta(\text{HTTHHTHTHH})$? Equivalently, which θ makes this observed sequence most likely to occur?

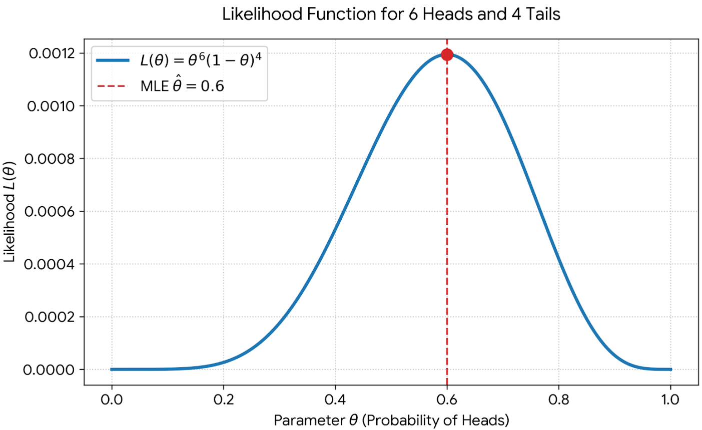
Likelihood

- This can be formally expressed as follows: Let $X_1, \ldots, X_{10}$ be independent Bernoulli random variables with parameter θ.
- The (joint) probability of observing the data $\boldsymbol{x} = (x_1, \ldots, x_{10})^T$, viewed as a function of the parameter θ, is defined as the *likelihood function* $L_{n=10}(\theta)$:

$$L_{n=10}(\theta) = \prod_{i=1}^{10} P_\theta(X_i = x_i) = \theta^{\sum_{i=1}^{10} x_i}(1-\theta)^{10-\sum_{i=1}^{n} x_i}. \tag{2}$$

- For our sequence with 6 heads and 4 tails, $L_n(\theta) = \theta^6(1-\theta)^4$.

Likelihood



- Finding the θ that maximizes this function is the core of Maximum Likelihood Estimation (MLE).

Outline

Maximum Likelihood Estimation

- The MLE is defined as:

$$\hat{\theta}_n^{\mathrm{MLE}} \in \arg\max_{\theta \in \Theta} L_n(\theta). \tag{3}$$

- Let $x_1, \ldots, x_n$ be independent samples from a Bernoulli distribution with parameter θ. Then, the likelihood function can be expressed as:

$$L_n(\theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^{n} x_i}(1-\theta)^{n-\sum_{i=1}^{n} x_i}. \tag{4}$$

- The logarithm of $L_n(\theta)$ is referred to as the *log-likelihood*. In this case,

$$l_n(\theta) = \log L_n(\theta) = \left(\sum_{i=1}^{n} x_i\right) \log\theta + \left(n - \sum_{i=1}^{n} x_i\right) \log(1-\theta). \tag{5}$$

Maximum Likelihood Estimation

- By differentiating with respect to θ, we obtain:

$$
\begin{aligned}
\frac{d}{d\theta} l_n(\theta) &= \frac{d}{d\theta}\left(\left(\sum_{i=1}^{n} x_i\right)\log\theta + \left(n - \sum_{i=1}^{n} x_i\right)\log(1-\theta)\right) \\
&= \frac{n\bar{x}}{\theta} - \frac{n - n\bar{x}}{1-\theta} > 0 \\
&= \frac{n}{\theta(1-\theta)}(\bar{x} - \theta) \iff \theta < \bar{x}.
\end{aligned} \tag{6}
$$

- For simplicity, assume $0 < \sum_{i=1}^{n} x_i < n$.[1] Then, the MLE is given by:

$$\hat{\theta}_n^{\mathrm{MLE}} = \bar{x}.$$

[1] Even if $\sum_{i=1}^{n} x_i = 0$ or n, the sample mean $\bar{x}$ still maximizes the (log-)likelihood.

Maximum Likelihood Estimation

- Now, let $x_1, \ldots, x_n$ be independent samples from a Gaussian distribution with parameter $\theta = (\mu, \sigma^2)^T$. Then, the likelihood function can be expressed as:

$$L_n(\theta) = \prod_{i=1}^{n} (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{2\sigma^2}\right). \tag{7}$$

- The log-likelihood is given by:

$$l_n(\theta) = \log L_n(\theta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2. \tag{8}$$

Maximum Likelihood Estimation

- Let's find the MLE for θ:

Maximum Likelihood Estimation

- Let θ_0 be the true (unknown) parameter value.
- Note that, in the Bernoulli case, the MLE $\hat{\theta}_n^{\text{MLE}} = \bar{x}$ enjoys the following properties:
  1. (Consistency) By the Law of Large Numbers, $\hat{\theta}_n^{\text{MLE}}$ converges in probability to θ_0:

$$P_{\theta_0}\left(|\hat{\theta}_n^{\text{MLE}} - \theta_0| > \epsilon\right) \to 0 \quad \text{as } n \to \infty. \tag{9}$$

  2. (Asymptotic Normality) By the Central Limit Theorem, for any z:

$$P_{\theta_0}\left(\frac{\hat{\theta}_n^{\text{MLE}} - \theta_0}{\sqrt{\text{Var}[\hat{\theta}_n^{\text{MLE}}]}} \le z\right) \to \Phi(z) := P(Z \le z) \quad \text{where } Z \sim N(0,1). \tag{10}$$

- These properties hold in general for MLEs under standard regularity conditions.

MLEs as Minimizers of Statistical Distances

- The MLE can also be justified as minimizing a statistical distance. We first introduce the Kullback-Leibler (KL) divergence.
- **Kullback-Leibler Divergence**: Let $f_1(x)$ and $f_2(x)$ be PDFs such that $\mathrm{supp}(f_1) \subseteq \mathrm{supp}(f_2)$. The KL-divergence of f_2 from f_1 is defined as:

$$\mathcal{D}_{\mathrm{KL}}(f_1 || f_2) = \int_{-\infty}^{\infty} f_1(x) \log\left(\frac{f_1(x)}{f_2(x)}\right) dx. \tag{11}$$

- Note that $g(t) = t \log t$ is a strictly convex function with $g(1) = 0$. The KL-divergence corresponds to the *f-divergence* (1) generated by this function g.

MLEs as Minimizers of Statistical Distances

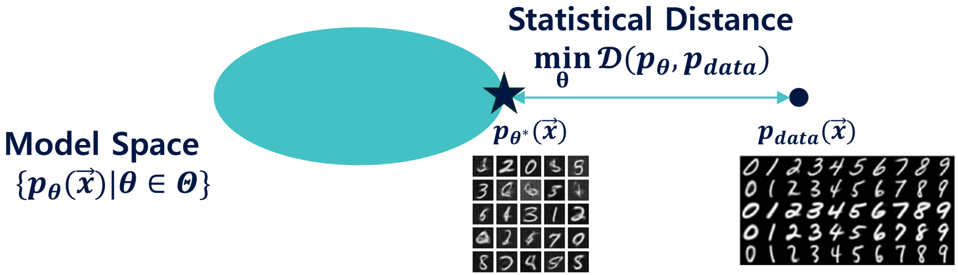
- As a valid divergence measure, the KL-divergence satisfies $\mathcal{D}_{\mathrm{KL}}(f_1||f_2) \geq 0$, and equality holds if and only if $f_1 = f_2$.

MLEs as Minimizers of Statistical Distances

- Let f_{data} be the data PDF. Then,

$$\begin{aligned} \mathcal{D}_{\mathrm{KL}}(f_{\text{data}} \parallel f_\theta) &= \int_{-\infty}^{\infty} \left(\log \frac{f_{\text{data}}(x)}{f_\theta(x)} \right) f_{\text{data}}(x)\, dx \\ &= -\int \left(\log f_\theta(x)\right) f_{\text{data}}(x) dx + \int \left(\log f_{\text{data}}(x)\right) f_{\text{data}}(x) dx \\ &\approx -\frac{1}{n}\sum_{i=1}^{n} \log f_\theta(X_i) + \text{Constant} \quad \text{where } X_1, \ldots X_n \overset{\text{i.i.d.}}{\sim} f_{\text{data}}(x) \\ &= -n^{-1} l_n(\theta) + \text{Constant}. \end{aligned} \tag{12}$$

MLEs as Minimizers of Statistical Distances



- Thus, maximizing the (log-)likelihood is equivalent to minimizing the empirical KL-divergence.

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Outline

Statistical Hypothesis Test

- Consider a scenario where we have two coins:
  1. **Coin 0**: A fair coin with a probability of .5 of getting heads.
  2. **Coin 1**: An unfair coin with a probability of .7 of getting heads.
- Suppose someone selects one of the two coins (with unknown probabilities for each coin) and flips the chosen coin 10 times, resulting in 6 heads.
- **Q**: Based on observing 6 heads, how can we determine whether the chosen coin is Coin 0 or Coin 1?

Bayesian Approach

- Consider two competing *hypotheses*:

$$H_0 : \text{Coin 0 (fair) was chosen} \quad \text{vs.} \quad H_1 : \text{Coin 1 (unfair) was chosen} \tag{13}$$

- Let $P(H_0)$ and $P(H_1)$ denote the prior probabilities (our initial degrees of belief) for each hypothesis. After observing the data, we can employ the posterior odds for decision-making:[2]

$$\frac{P(H_0 \mid \text{Data})}{P(H_1 \mid \text{Data})}. \tag{14}$$

A higher value indicates stronger evidence in favor of H_0 relative to H_1.

[2] Recall that the odds of an event A are defined as $P(A)/P(A^c)$.

Bayesian Approach

- Let x denote the observed number of heads. Note that Bayes' rule implies:

$$\underbrace{\frac{P(H_0 \mid x)}{P(H_1 \mid x)}}_{\text{Posterior Odds}} = \underbrace{\frac{P(H_0)}{P(H_1)}}_{\text{Prior Odds}} \underbrace{\frac{P(x \mid H_0)}{P(x \mid H_1)}}_{\text{Likelihood Ratio}} . \tag{15}$$

  This implies that for any threshold constant c',

$$\underbrace{\frac{P(H_0 \mid x)}{P(H_1 \mid x)}}_{\text{Posterior Odds}} > c' \iff \underbrace{\frac{P(x \mid H_0)}{P(x \mid H_1)}}_{\text{Likelihood Ratio}} > c := c' \Bigg(\underbrace{\frac{P(H_0)}{P(H_1)}}_{\text{Prior Odds}} \Bigg)^{-1} . \tag{16}$$

- Therefore, a higher likelihood ratio implies higher posterior odds. We accept H_0 if the evidence is strong enough to outweigh the prior belief.

Bayesian Approach

x	0	1	2	3	4	5	6	7	8	9	10
coin 0	.0010	.0098	.0439	.1172	.2051	.2461	.2051	.1172	.0439	.0098	.0010
coin 1	.0000	.0001	.0014	.0090	.0368	.1029	.2001	.2668	.2335	.1211	.0282

x	0	1	2	3	4	5	6	7	8	9	10
$\frac{P(x\|H_0)}{P(x\|H_1)}$	165.4	70.88	30.38	13.02	5.579	2.391	1.025	0.4392	0.1882	0.0807	0.0346

Fig. 9.1 (top) and Fig. 9.2 (bottom) from MSDA

Bayesian Approach

- Suppose $P(H_0)/P(H_1) = 1$ and we decide to accept H_0 if:

$$\frac{P(H_0 \mid x)}{P(H_1 \mid x)} > 1 \iff \frac{P(x \mid H_0)}{P(x \mid H_1)} > \left(\frac{P(H_0)}{P(H_1)}\right)^{-1} = 1. \tag{17}$$

- The probability of accepting or rejecting H_0 depends on what the true coin was.
- For example, if the chosen coin is Coin 0 (fair):

$$P(\text{accept } H_0 \mid H_0) = P\left(\left\{x : \frac{P(x \mid H_0)}{P(x \mid H_1)} > 1\right\} \mid H_0\right) = P(X < 7 \mid H_0) = .82. \tag{18}$$

Consequently, the probability of making an incorrect decision (rejecting H_0 when it is true) is $P(\text{reject } H_0 \mid H_0) = .18$.

Bayesian Approach

- In contrast, if the chosen coin is Coin 1 (unfair):

$$P(\text{reject } H_0 \mid H_1) = P\left(\left\{x : \frac{P(x \mid H_0)}{P(x \mid H_1)} \leq 1\right\} \mid H_1\right) = P(X \geq 7 \mid H_1) = .65. \tag{19}$$

Consequently, the probability of making another incorrect decision (accepting H_0 when it is not true) is $P(\text{accept } H_0 \mid H_1) = .35$.

The Neyman-Pearson Paradigm

		Truth (State of Nature)	
		H_0 is True	H_0 is False
Decision	Accept H_0	Correct	Type II Error (β)
	Reject H_0	Type I Error (α)	Correct

- The *Neyman-Pearson paradigm* views the truth of a hypothesis as a fixed state of nature, establishing the optimality of likelihood-ratio-based decision rules.
- This paradigm aims to find a decision rule that maximizes the *power*, $1 - \beta = P(\text{reject } H_0 \mid H_1)$, while controlling the *significance level*, $\alpha = P(\text{reject } H_0 \mid H_0)$.

The Neyman-Pearson Paradigm

- Let $X_1, \ldots, X_n$ be independent samples from $\mathcal{N}(\mu, 1)$ with unknown $\theta = \mu$. Suppose we test two competing hypotheses:

$$H_0 : \theta = \theta_0 = 0 \quad \text{vs.} \quad H_1 : \theta = \theta_1 = 1. \tag{20}$$

- For a given threshold $c > 0$, the *likelihood ratio test* rejects H_0 if:

$$\frac{f(\boldsymbol{x} \mid H_1)}{f(\boldsymbol{x} \mid H_0)} = \frac{f_{\theta_1}(\boldsymbol{x})}{f_{\theta_0}(\boldsymbol{x})} = \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i - 1)^2\right)}{\exp\left(-\frac{1}{2}\sum_{i=1}^{n} x_i^2\right)} = \exp\left(\sum_{i=1}^{n} x_i - \frac{n}{2}\right) > c. \tag{21}$$

  This condition is equivalent to $\bar{x} > c'$ for some constant c'.
- Such decision rules can be formulated as:

$$d^{\mathrm{LR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad \bar{X} > c' \\ 0 & \text{if} \quad \bar{X} \leq c'. \end{cases} \tag{22}$$

The Neyman-Pearson Paradigm

- A naturally arising question is whether there exists a more powerful test than the likelihood ratio test at a fixed level of significance α.
- <u>**Most Powerful Test**</u>: Given $\boldsymbol{X} = (X_1, \ldots, X_n)^T$ from a distribution with PDF $f_\theta(\boldsymbol{x})$[3], suppose we test two hypotheses:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1. \tag{23}$$

A test $d^{\mathrm{MP}}(\boldsymbol{X}) \in \{0, 1\}$ is defined as the *most powerful test* of level $0 < \alpha < 1$ if it satisfies the following conditions:

1. (Type I error) $\mathrm{E}_{\theta_0}[d^{\mathrm{MP}}(\boldsymbol{X})] \leq \alpha$,
2. (Power) $\mathrm{E}_{\theta_1}[d^{\mathrm{MP}}(\boldsymbol{X})] \geq \mathrm{E}_{\theta_1}[d(\boldsymbol{X})]$ for any test $d(\boldsymbol{X})$ such that $\mathrm{E}_{\theta_0}[d(\boldsymbol{X})] \leq \alpha$.

[3] This definition generalizes naturally to discrete distributions as well.

The Neyman-Pearson Paradigm

- **The Neyman-Pearson Lemma**: Suppose we test two hypotheses:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1. \tag{24}$$

For a given significance level α, if a *likelihood ratio test* $d^{\mathrm{LR}}(\boldsymbol{X})$ defined as

$$d^{\mathrm{LR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad f_{\theta_1}(\boldsymbol{X})/f_{\theta_0}(\boldsymbol{X}) > c \\ 0 & \text{if} \quad f_{\theta_1}(\boldsymbol{X})/f_{\theta_0}(\boldsymbol{X}) \leq c \end{cases} \tag{25}$$

satisfies $\mathrm{E}_{\theta_0}[d^{\mathrm{LR}}(\boldsymbol{X})] = \alpha$ for some constants $c \geq 0$, then it is the most powerful test of level α.

Generalized Likelihood Ratio Test

- Though the Neyman-Pearson paradigm concretely justifies the optimality of LR tests, it faces limitations when the hypotheses become composite.
- For example, let $X_1, \ldots, X_n$ be independent samples from $\mathcal{N}(\mu, 1)$ with unknown $\theta = \mu$, and now we test two competing hypotheses:

$$H_0 : \theta = \theta_0 = 0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0. \tag{26}$$

- Now the alternative includes multiple choices of f_θ. It is known that there is no test that is uniformly more powerful than all other tests in this setting.
- The *generalized likelihood ratio test* provides a practical alternative, and is defined by the decision rule:

$$d^{\mathrm{GLR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad \frac{\sup_{\theta \in \Theta} f_\theta(\boldsymbol{X})}{\sup_{\theta \in \Theta_0} f_\theta(\boldsymbol{X})} > c \\ 0 & \text{if} \quad \frac{\sup_{\theta \in \Theta} f_\theta(\boldsymbol{X})}{\sup_{\theta \in \Theta_0} f_\theta(\boldsymbol{X})} \leq c \end{cases} \tag{27}$$

Generalized Likelihood Ratio Test

- To derive the GLR test, we first find the MLEs under the two parameter spaces:
  - Under $\Theta_0 = \{0\}$: The restricted maximum is simply $L(0)$.
  - Under $\Theta = \mathbb{R}$: The unrestricted MLE is $\hat{\theta}^{\text{MLE}} = \bar{X}$, giving $L(\bar{X})$.
- The GLR statistic is constructed as:

$$\frac{\sup_{\theta \in \Theta} L(\theta)}{\sup_{\theta \in \Theta_0} L(\theta)} = \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^{n}(X_i - \bar{X})^2\right)}{\exp\left(-\frac{1}{2}\sum_{i=1}^{n} X_i^2\right)} > c. \tag{28}$$

  This condition is equivalent to $|\bar{X}| > c'$ for some constant c'.
- The GLR test at significance level α can be formulated as:

$$d^{\text{GLR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad |\bar{X}| > z_{\alpha/2}/\sqrt{n} \\ 0 & \text{if} \quad |\bar{X}| \le z_{\alpha/2}/\sqrt{n}. \end{cases} \tag{29}$$

Outline

Two-sample Comparison

- In the previous subsection, we studied hypothesis testing for a single population.
- Now, we consider cases where we have two groups of samples, e.g., synthetic images from generative models and real images.
- For simplicity, we assume that populations of both groups are Gaussian with unit variance. We can formulate the data as follows:

$$\begin{aligned} \text{Group 1}: \quad & X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_X, 1) \\ \text{Group 2}: \quad & Y_1, \ldots, Y_m \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_Y, 1). \end{aligned} \tag{30}$$

- We want to test whether the two populations share the same mean:

$$H_0 : \mu_X = \mu_Y = \mu \quad \text{vs.} \quad H_1 : \mu_X \neq \mu_Y. \tag{31}$$

Two-sample Comparison

- To derive the GLR test, we first find the MLEs under the two parameter spaces:
  - Under Θ_0 ($\mu_X = \mu_Y = \mu$): The restricted MLE for the common mean is the pooled sample mean, $\hat{\mu}_0 = \frac{\sum X_i + \sum Y_j}{n+m} = \frac{n\bar{X} + m\bar{Y}}{n+m}$.
  - Under $\Theta = \mathbb{R}^2$: The unrestricted MLEs are $\hat{\mu}_X = \bar{X}$ and $\hat{\mu}_Y = \bar{Y}$.
- The GLR statistic is constructed by taking the ratio of the supremums:

$$\frac{\exp\left(-\frac{1}{2}\sum_{i=1}^{n}(X_i - \bar{X})^2 - \frac{1}{2}\sum_{j=1}^{m}(Y_j - \bar{Y})^2\right)}{\exp\left(-\frac{1}{2}\sum_{i=1}^{n}(X_i - \hat{\mu}_0)^2 - \frac{1}{2}\sum_{j=1}^{m}(Y_j - \hat{\mu}_0)^2\right)} > c. \tag{32}$$

This condition can be shown to be equivalent to:

$$|\bar{X} - \bar{Y}| > c'. \tag{33}$$

Two-sample Comparison

- When the variance is known ($\sigma^2 = 1$), under H_0, the difference follows $\bar{X} - \bar{Y} \sim \mathcal{N}\left(0, \frac{1}{n} + \frac{1}{m}\right)$. The GLR test at significance level α is:

$$d^{\text{GLR}}(\boldsymbol{X}, \boldsymbol{Y}) = \begin{cases} 1 & \text{if} \quad |\bar{X} - \bar{Y}| > z_{\alpha/2}\sqrt{\frac{1}{n} + \frac{1}{m}} \\ 0 & \text{if} \quad |\bar{X} - \bar{Y}| \leq z_{\alpha/2}\sqrt{\frac{1}{n} + \frac{1}{m}}. \end{cases} \tag{34}$$

- When we further assume that the population variance σ^2 is unknown (but equal for both groups), the GLR test leads to the *two-sample Student's t-test*.
- The test statistic replaces the true variance with the pooled sample variance s_p^2:

$$T = \frac{\bar{X} - \bar{Y}}{s_p\sqrt{\frac{1}{n} + \frac{1}{m}}}, \quad \text{where} \quad s_p^2 = \frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2}. \tag{35}$$

Under H_0, $T \sim t_{n+m-2}$, leading to the rejection region $|T| > t_{\alpha/2, n+m-2}$.

Outline

The Trinity of Hypothesis Tests

- In this subsection, we illustrate the geometric intuition behind the GLR test and its fundamental connections to other test statistics. We first define the *score function* and *Fisher information*.
- **Score Function**: The *score function* $s_n(\theta)$ is defined as the gradient (slope) of the log-likelihood with respect to θ:

$$s_n(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \log f(X_i; \theta) = l_n'(\theta). \tag{36}$$

- **Fisher Information**: The *Fisher Information* $\mathcal{I}_n(\theta)$ is defined as the variance of the score function:

$$\mathcal{I}_n(\theta) = \text{Var}_\theta[s_n(\theta)]. \tag{37}$$

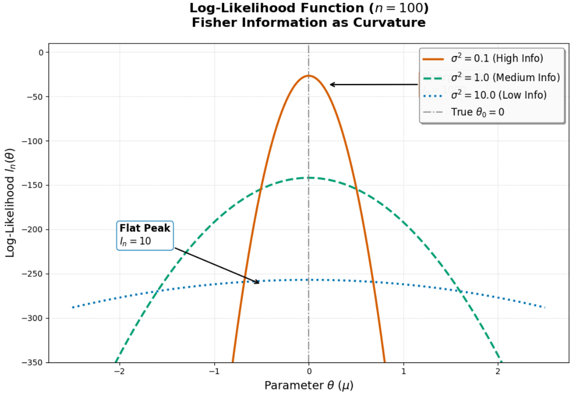
The Trinity of Hypothesis Tests

- Let's evaluate these quantities for $X \sim \mathcal{N}(\mu, \sigma^2)$, where $\theta = \mu$ (estimating the mean with a known variance). The log-likelihood for a single observation is $l_1(\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(X-\mu)^2}{2\sigma^2}$.
- The score function is $s_1(\mu) = l_1'(\mu) = \frac{X-\mu}{\sigma^2}$. Note that the expected score is zero:

$$\mathrm{E}_\mu[s_1(\mu)] = \mathrm{E}_\mu\left[\frac{X-\mu}{\sigma^2}\right] = 0. \tag{38}$$

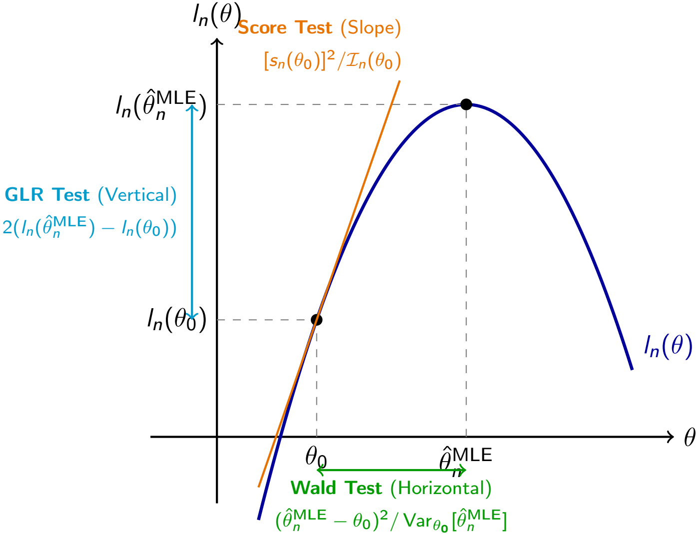
- Taking the second derivative yields $l_1''(\mu) = -\frac{1}{\sigma^2}$. Thus, the Fisher information is $\mathcal{I}_1(\mu) = -\mathrm{E}_\mu[l_1''(\mu)] = \frac{1}{\sigma^2}$.
- The identities $\mathrm{E}_\theta[s_1(\theta)] = 0$ and $\mathcal{I}_1(\theta) = -\mathrm{E}_\theta[l_1''(\theta)]$ hold in general under standard regularity conditions.

The Trinity of Hypothesis Tests



- The relation $\mathcal{I}_1(\theta) = -\mathrm{E}_\theta[l_1''(\theta)]$ represents the expected curvature of the log-likelihood $l_n(\theta)$.
- For $\mathcal{N}(\mu, \sigma^2)$, a larger Fisher information ($\mathcal{I}_1(\mu) = \frac{1}{\sigma^2}$) indicates a smaller variance σ^2.
- Geometrically, this means that the log-likelihood curve has a sharper peak, indicating that the data contains more precise information about the parameter μ.

The Trinity of Hypothesis Tests



The Trinity of Hypothesis Tests

- **The Trinity of Hypothesis Tests**: Under the null hypothesis $H_0 : \theta = \theta_0$, all three foundational tests, GLR, Wald, and Rao's score, are asymptotically equivalent.
- Specifically, as the sample size $n \to \infty$, all three test statistics converge in distribution to χ^2_1.
- This remarkable property implies that for large samples, regardless of whether we measure the horizontal distance (Wald), the vertical drop (GLR), or the slope (Score), we will reach the exact same statistical conclusion.

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